

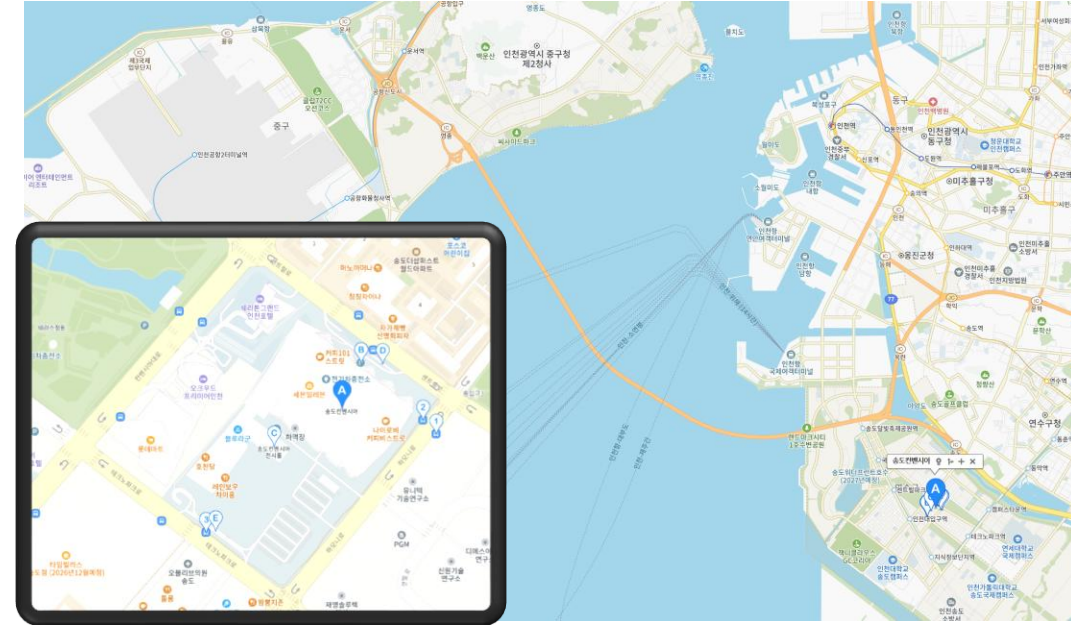
# 29<sup>th</sup> Roboracer Autonomous Racing Competition Orientation

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2026, August 24<sup>th</sup> ~ 27<sup>th</sup>

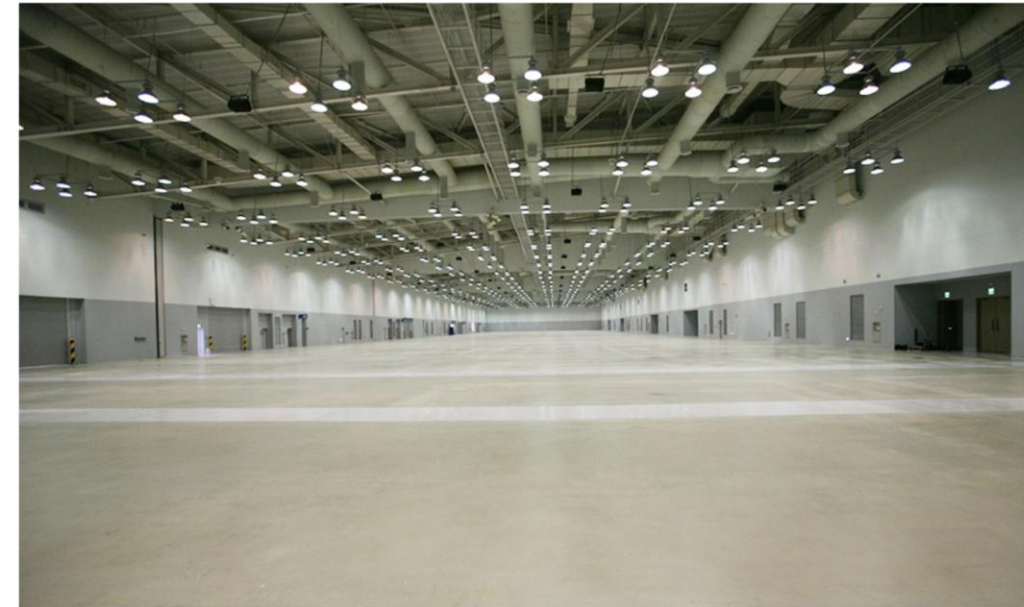
# Competition Overview

- IFAC (International Federation of Automatic Control)
  - August 23(Sun)~ August 28(Fri), 2026
  - **Roboracer Competition:**  
**August 24(Mon)~ August 27(Thu), 2026**
  - Bexco, Busan, South Korea



# Projected Competition Place

## BEXCO Exhibition Center 2, 3<sup>rd</sup> floor Hall 5A



Hall 5 (3F)

Slippery Surface

# Projected Competition Place

## Roboracer Competition Area





# Homepage Info.

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- The IFAC 2026 Roboracer homepage is open
  - <https://2026ifac-robocer.com/en/registration.html>
- Competition information, rules, and registration (~8.9<sup>th</sup>) are available through the website.



# Registration(~8.9<sup>th</sup>)

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## REGISTRATION

1

### Conference Registration

Register for IFAC 2026.  
See the fee table for  
registration costs.

**REGISTER FOR IFAC**

2

### Submit Google Form

Enroll your team information.  
We manage the competition  
based on this.

**SUBMIT FORM**

3

### Confirm Registration

Check the participant list  
below  
to verify your team is  
registered.

**VIEW PARTICIPANTS**

If you completed steps 1 & 2 but your team is not listed, please contact [ryujs@unist.ac.kr](mailto:ryujs@unist.ac.kr).

## You can submit google form first

# Registration(~8.9<sup>th</sup>)

- **Conference Registration**

- [https://ifac2026.org/fairContents.do?FAIRMENU\\_IDX=21697&hl=ENG](https://ifac2026.org/fairContents.do?FAIRMENU_IDX=21697&hl=ENG)

- Every team member must individually register for IFAC — not just the team leader.
- If you have already registered for IFAC under any category, you do not need to add the competition option separately.

## REGISTRATION

|                                                                                                                                                         |                                                                                                                                                  |                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br><b>Conference Registration</b><br>Register for IFAC 2026.<br>See the fee table for registration costs.<br><a href="#">REGISTER FOR IFAC</a> | <b>2</b><br><b>Submit Google Form</b><br>Enter your team information.<br>We manage the competition based on this.<br><a href="#">SUBMIT FORM</a> | <b>3</b><br><b>Confirm Registration</b><br>Check the participant list below to verify your team is registered.<br><a href="#">VIEW PARTICIPANTS</a> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

If you completed steps 1 & 2 but your team is not listed, please contact ryujs@unist.ac.kr.

### ON-LINE REGISTRATION (FEB. 1 - AUG. 9, 2026)

€ 100

### ON-SITE REGISTRATION (AUG. 22 - 28, 2026)

€ 140

# Registration(~8.9<sup>th</sup>)

- **Competition Google Form Submission**

- <https://docs.google.com/forms/d/e/1FAIpQLSdJg-bRTA8269yDE7EInUQxiU9iTom4lwIJWPxlzpofVzsc8A/viewform>
- **No additional charge**
- Submission per team
- Hardware spec, brief algorithm description and etc.

## 29th Roboracer Autonomous Racing Competition

**Roboracer Autonomous Racing** is a semi-regular competition organized by an international community of researchers, engineers, and autonomous systems enthusiasts. The teams participating in **29th Roboracer CHAMPIONSHIP** will build a 1:10 scaled autonomous race car according to a given specification and write software for it to fulfill the objectives for the competition: Don't crash and minimize the lap time.

The registration for the 2026 29th Roboracer autonomous racing competition is **26.06.30**

Upon completion of the registration process, it is mandatory to **join the official Slack channel**.

Any inquires please contact to email below  
ryujs@unist.ac.kr

### REGISTRATION

|                                                                                                                                                                       |                                                                                                                                                             |                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>1</div> <div>Conference<br/>Registration</div> <div>Register for IFAC 2026.<br/>See the fee table for<br/>registration costs.</div> <div>REGISTER FOR IFAC</div> | <div>2</div> <div>Submit Google Form</div> <div>Enroll your team information.<br/>We manage the competition<br/>based on this.</div> <div>SUBMIT FORM</div> | <div>3</div> <div>Confirm<br/>Registration</div> <div>Check the participant list<br/>below<br/>to verify your team is<br/>registered.</div> <div>VIEW PARTICIPANTS</div> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

If you completed steps 1 & 2 but your team is not listed, please contact ryujs@unist.ac.kr.



# Registration(~8.9<sup>th</sup>)

- Confirm your status
  - <https://2026ifac-roboracer.com/en/registration.html#participants>

1

Conference  
Registration

Register for IFAC 2026.  
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REGISTER FOR IFAC

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Check the participant list  
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VIEW PARTICIPANTS

If you completed steps 1 & 2 but your team is not listed, please contact ryuj@unist.ac.kr.

## PARTICIPANTS

If you have registered for participation but the list below is not updated, please contact us at ryuj@unist.ac.kr.

| TEAM NAME | AFFILIATION | TEAM MEMBERS |
|-----------|-------------|--------------|
| .         | .           | .            |

# Tentative Timeline

| DATE                               | IN-PERSON COMPETITION                                  |
|------------------------------------|--------------------------------------------------------|
| Feb 1 <sup>st</sup>                | Registration opens                                     |
| June 12 <sup>th</sup><br>20:00 KST | 1 <sup>st</sup> Orientation (via Online)               |
| Late July                          | 2 <sup>nd</sup> Orientation (via Online)               |
| Aug 9 <sup>th</sup><br>23:59 KST   | Registration closes                                    |
| Aug 24 <sup>th</sup>               | On-site team reception and regulated practice sessions |
| Aug 25 <sup>th</sup>               | Open practice sessions                                 |
| Aug 26 <sup>th</sup>               | Qualification time trials                              |
| Aug 27 <sup>th</sup>               | Head-to-head tournament and award ceremony             |



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# Tentative Competition Schedule

| Timeline | Day1                 |
|----------|----------------------|
| 10:00    | Registration         |
| 12:00    | Mapping & Inspection |
| 15:00    | Group Practice       |
| 18:00    | Open Practice        |
| 21:00    | Close Track          |

| Timeline | Day2           |
|----------|----------------|
| 10:00    | Group Practice |
| 15:00    | Qualifying     |
| 18:00    | Open Practice  |
| 21:00    | Close Track    |

| Timeline | Day3                       |
|----------|----------------------------|
| 10:00    | Group Practice             |
| 14:00    | Half of head-to-head races |
| 18:00    | Open Practice              |
| 21:00    | Close Track                |

| Timeline | Day4                       |
|----------|----------------------------|
| 10:00    | Group Practice             |
| 11:00    | Half of head-to-head races |
| 16:00    | Ceremony & Workshop        |

**Under the double elimination format, no teams will be eliminated on the third day. The losers' bracket will start on the Day 4.**



# Example of tournament Bracket



**Winners Bracket**



**Losers Bracket**

# Example of tournament Bracket

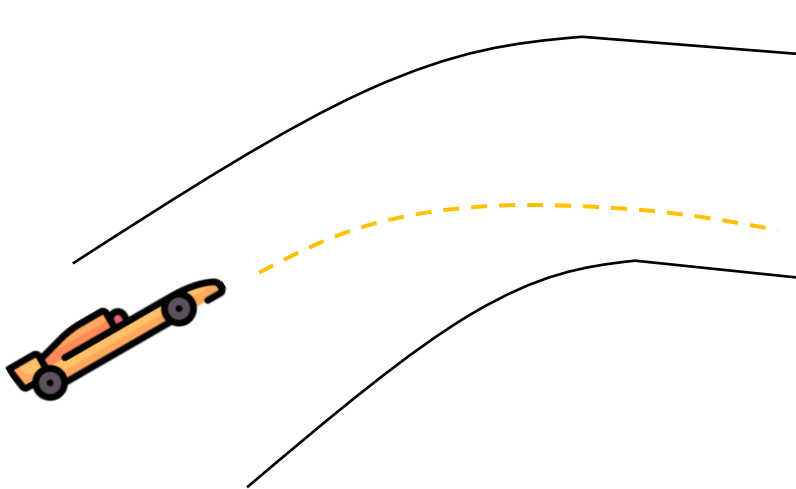


# RoboRacer Theme

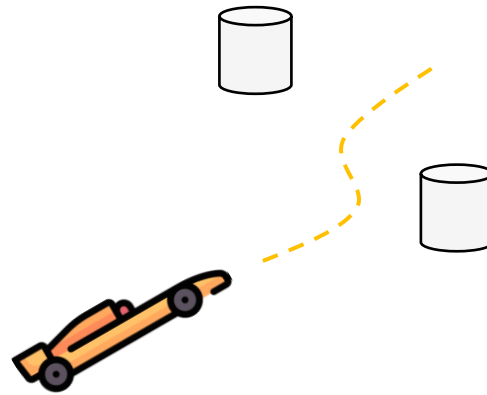
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# RoboRacer Theme

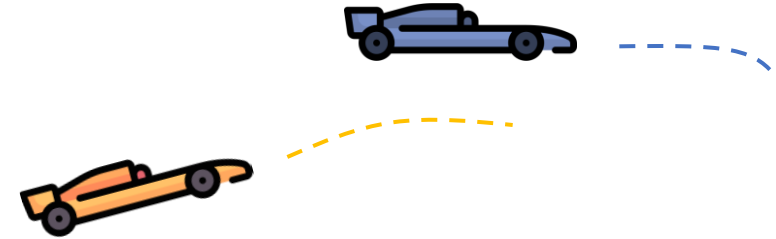
- Safer! Faster! More Precise!



Minimum Lap Time



Obstacle Avoidance



Adversarial Strategy

**Full Autonomous Racing System (Perception/Planning/Control)**

# RoboRacer Theme

- Key changes & challenges in the competition

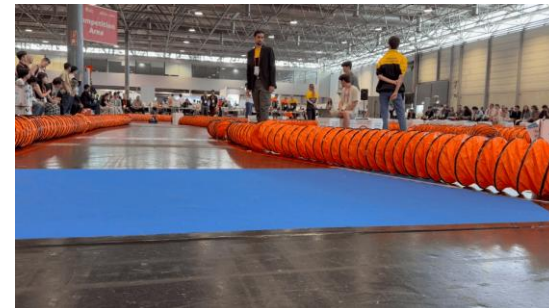
- Manual stop by a human to avoid collision (ex. Use joystick)
- Race focused on the optimal path
- Parameter tuning during the race



- No manual intervention during race (Fully autonomous system)
- Opponent detection, avoidance and interaction is more emphasized → precise decision-making is now crucial
- Parameter adjustments allowed only when the car is stationary

# Recent Competitions

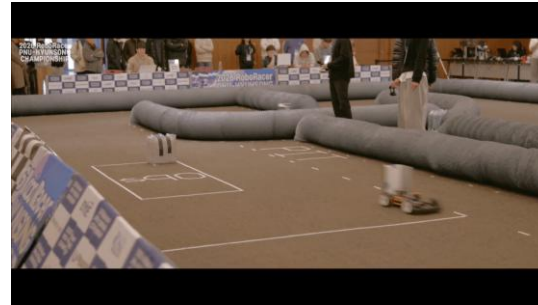
- 2026 PNU-HYUNSONG (01.25-01.27, Busan, Korea)
- 2026 ICRA (06.01-06.04, Vienna, Austria)





# Recent Competitions

- 2026 PNU-HYUNSONG (01.25-01.27, Busan, Korea)



## Similar Rules (Recommend to watch below videos)

### • Highlights

- <https://youtu.be/fWzsDwl2cmA?t=209>

### • Round of 16:

- <https://www.youtube.com/watch?v=Bjk7JBISy9Y&t=279s>

### • Round of 8:

- <https://www.youtube.com/watch?v=hciQyJ6drEw&t=33s>

# Phases Overview

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# Phases Overview

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## 0. Submission of Technical Materials

## 1. Registration & Car Inspection

## 3. Qualification

- Divide into Q1, Q2, Q3

## 2. Mapping & Practice Session

- Group Mapping
- Group Practice
- Open Practice

## 4. Head-to-head

- Standing Start
- Finish 20 laps before the opponent
- Unknown random obstacles

# Phase 1: Car Inspection

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- **Software inspection (Emergency Stop)**
  - Make **on-off toggle** for mode switch (Autonomous <-> Human Control)
  - When the car is autonomous mode, **do not hold any button** (ex. Press and hold button)



# Phase 1: Car Inspection

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- **Hardware Inspection**
- The car is supposedly built according to ROBORACER instructions, but alternative components may be allowed as long as they comply with the **Rules**.
  - Any unclear or ambiguous points should be checked **in advance** with the race organizers.
- The dedicated time slots for the Mapping and Group practice sessions will be designated on a **first-come-first-serve basis**, and only teams that have successfully **completed registration** and **inspection** will be eligible.

# Phase 1: Car Inspection

- **Chassis**

- The race is designed with 1:10 Traxxas chassis (e.g., TRA74054, TRA6804R)
- These chassis are recommended, but generally, any chassis within 15% of the Traxxas cars' dimensions is allowed ( $238\text{mm} \leq \text{Width} \leq 341\text{mm}$ ,  $454\text{ mm} \leq \text{Length} \leq 654\text{mm}$ ).
- Both 4WD and 2WD are permitted.

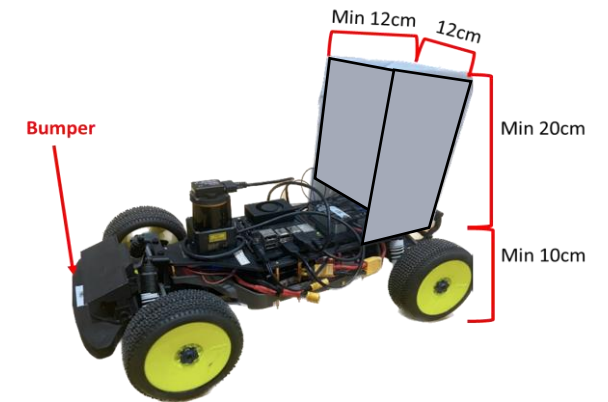
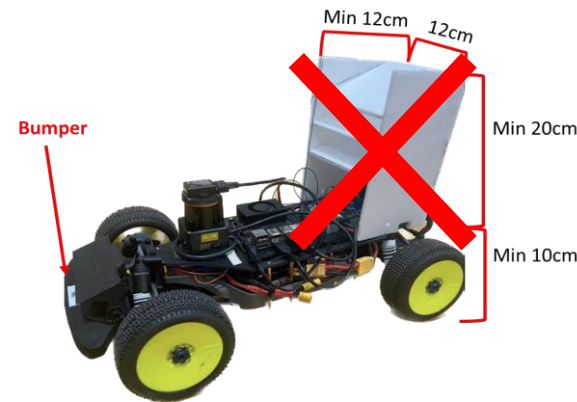
- **Tires**

- No limitations (Both sponge and rubber are allowed)
- However, **chemical additives are strictly prohibited.**



# Phase 1: Car Inspection

- **Detection Box (frequently violated and claimed from other teams)**
  - The car has to be easily perceivable by the opponent's LiDAR and/or Camera.
  - Therefore, the car must occupy a space of size **at least** 12×12 cm at **every horizontal plane**(4 sides, not 3 sides) between **10 to 30 cm** above the ground.
  - Ex. From ground 9~31cm(**O**), 11~31cm(**X**) , 9~29cm(**X**)
- **Soft Bumper**
  - It has to be soft to minimize the damage
  - Both **foam** or **honeycomb** are allowed
  - **Plastic bumper is not allowed**





# Phase 1: Car Inspection

- **Motor**

- No Limitation on spec.
- Only single motor can be used in power-train

- **Battery**

- 4s LiPo Battery or lower
  - 3S or lower per battery, and the total voltage must not exceed 4S
- No limit on number of batteries as long as total cell count in series does not exceed 4S
  - Ex) 2s + 2s (series), 4s+4s (parallel) are allowed,



# Phase 1: Car Inspection

- **Main Computation Unit**
  - Only 1 unit
  - No Limitations
    - E.g. Jetson Nano, Jetson Orin, Asus NUC



# Phase 1: Car Inspection

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- **LiDAR**

- No Limitations, but **only 1 LiDAR** can be used.
- In particular, **3D LiDAR** is also permitted (however, please be aware that high-cost 3D LiDAR units may be damaged in high-speed race).



# Phase 1: Car Inspection

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- **Camera**

- Both **mono camera** (e.g. Logitech C270, Logitech C920, Raspberry Pi Camera Module V2, Arducam) and **stereo cameras** (e.g. Intel Realsense, ZED) are allowed.
- Cameras that provide additional information such as **detection or VIO results** from onboard processing in camera are **not allowed**. (Depth information is allowed)

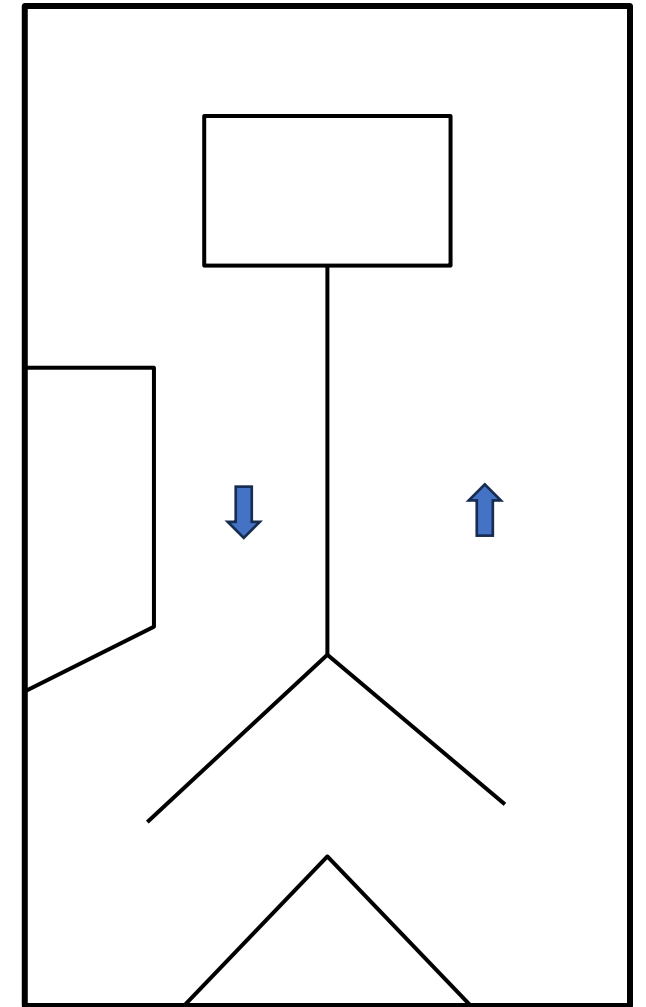
- **Other Sensors**

- Other sensors (IMUs, encoders, custom electronic speed controllers) are **allowed**.
- Indoor GPS sensors (e.g. Marvelmind) **are not allowed**.

# Phase 2: Mapping

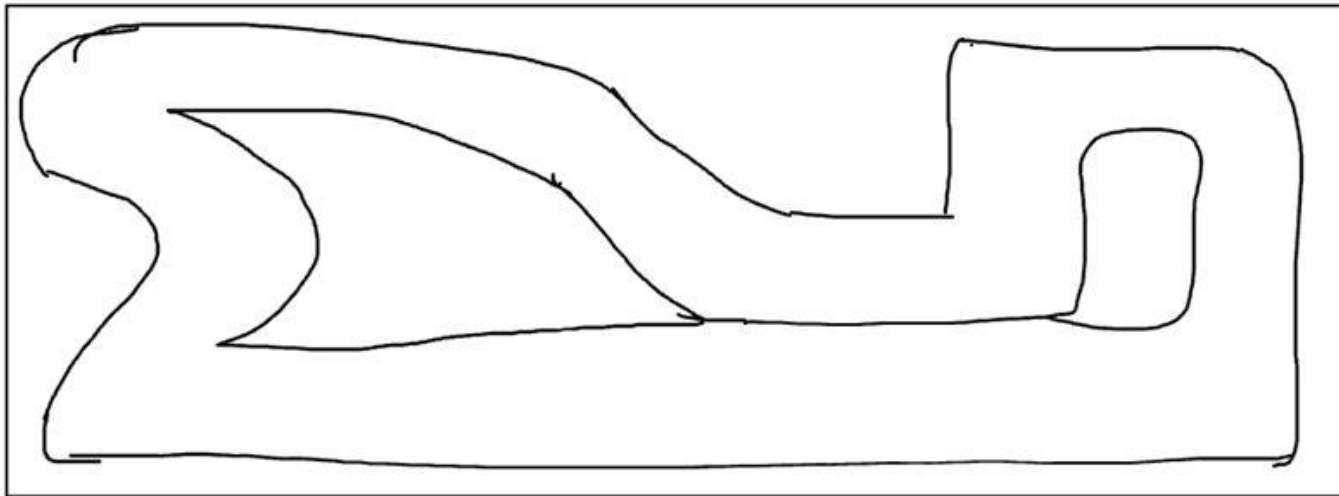
- **Mapping**

- Teams are divided into groups (about **3 groups**), and **30 minutes** is given to each group
- The **number of groups may vary** depending on the number of participating teams
- In this session, high speed driving and algorithm testing are **not** permitted.
  - This will be supervised by referees.
- **Do not make any crash at this session!**
- Multiple cars from a single team on the track is **not** permitted.
- The organizers will provide **track map files (.png, .yaml and .pbstream)**, but do not guarantee the quality.



# Phase 2: Mapping

- **Track Information**
  - Minimum Width: 1m



20m

8m

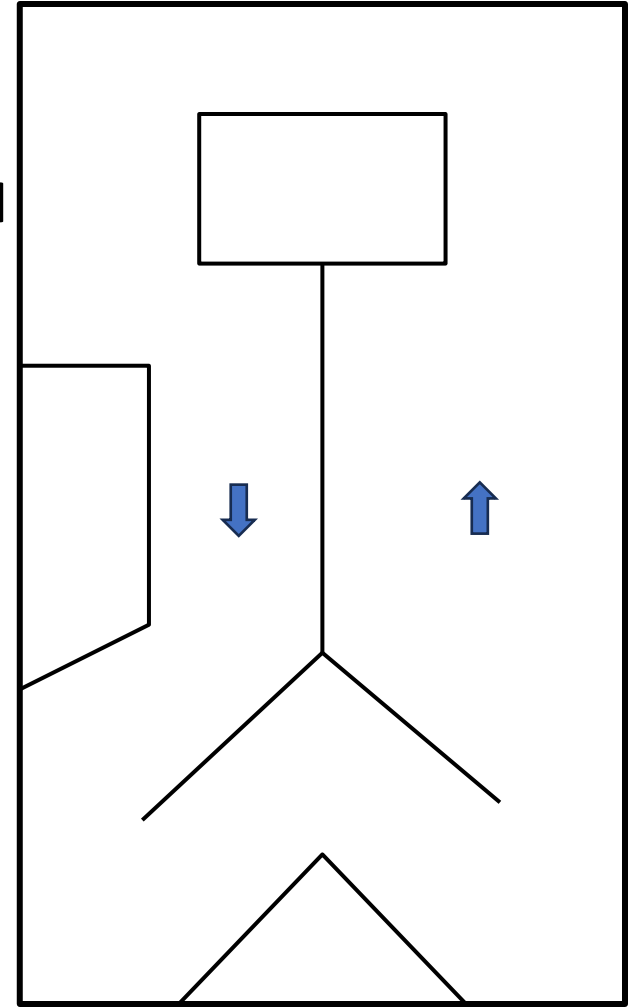


Gray Duct

# Phase 2: Practice

- **Practice**

- There will be a **regulated group practice** and **open practice for all team**
- The organizers will provide **sample obstacles**
- The organizers assume **no responsibility** for accidents during practice
- The team involved in an accident must **explain its algorithm** at the request of the referees





# Phase 3: Qualifying

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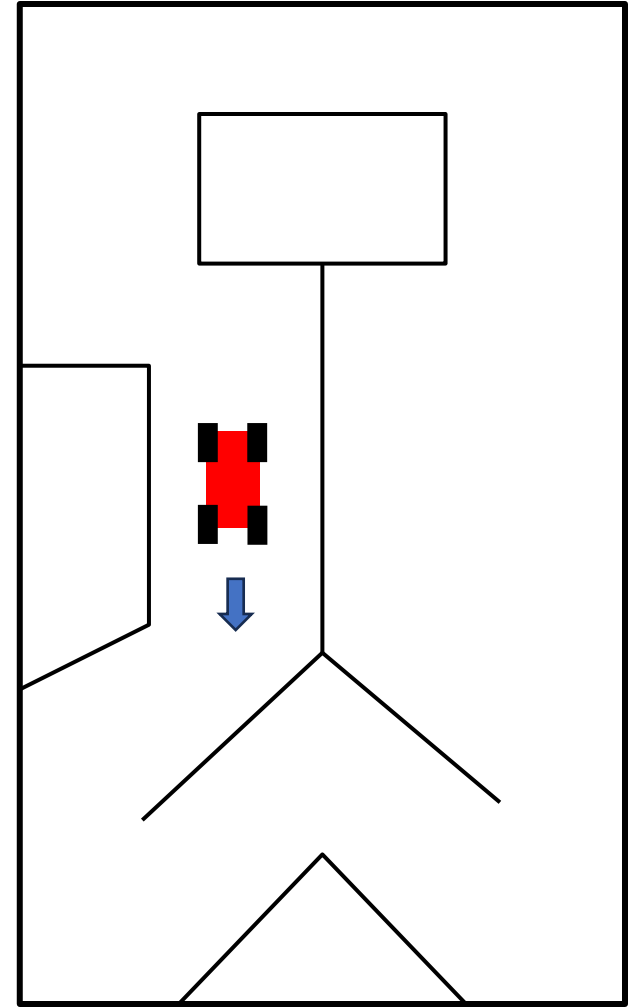
# Phase 3: Qualifying

- **Objective**

- Q1 — Complete 3 laps without collision at obstacle-free track
- Q2 — Complete 3 laps without collision while avoiding random obstacles
- Q3 — Achieve the minimum lap time on an obstacle-free track for 2 minutes
- Fully Autonomous (Achievement Condition)

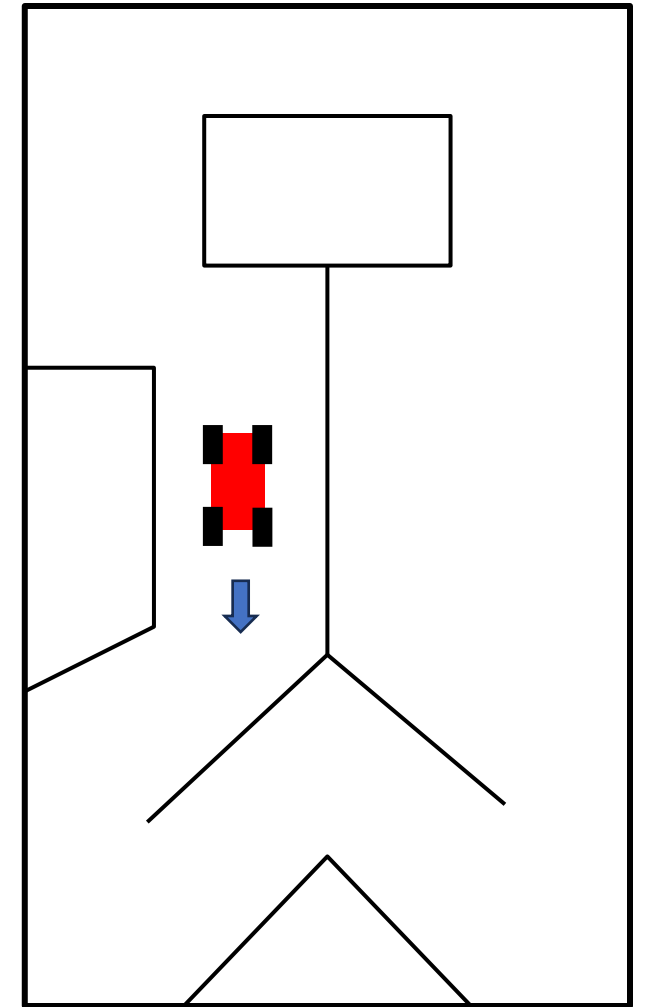
- **General**

- Both practice and qualifying use the same track
- Each team will have 8 minutes of time slot for record.
  - 8 minutes can differ depending on # of teams.



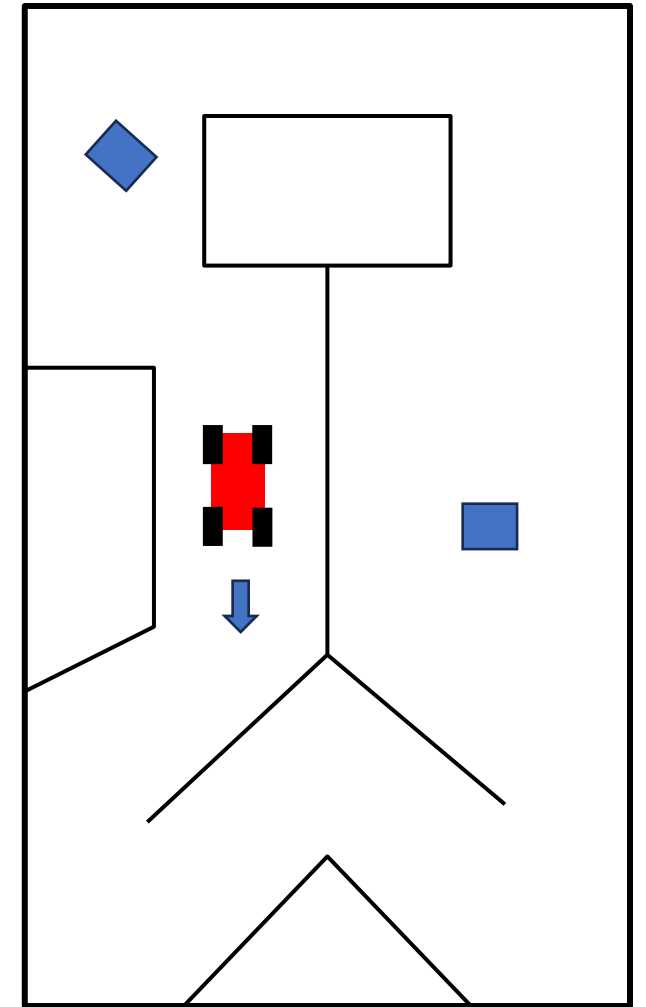
# Phase 3: Qualifying

- **Q1 — Complete 3 laps without collision at obstacle-free track**
  - If not passed, ranked by number of laps completed without collision
  - Passing → advance to Q2



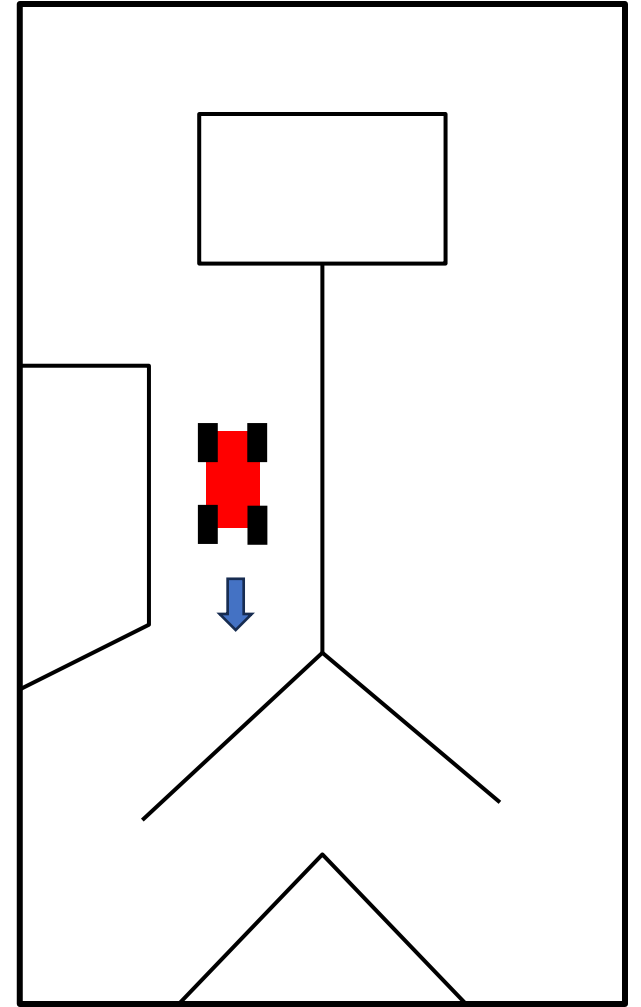
# Phase 3: Qualifying

- **Q2 — Complete 3 laps without collision while avoiding random obstacles**
  - 2 random obstacles placed by the referee (positions not announced in advance)
  - If not passed, ranked by number of laps completed without collision
  - Passing → advance to Q3



## Phase 3: Qualifying

- **Q3 — Achieve the minimum lap time on an obstacle-free track for 2 minutes**
  - The shortest lap time during the free 2-minute run is used



# Phase 3: Qualifying

- **Fully Autonomous (Achievement Condition)**
  - Applies at every moment throughout qualifying
  - Condition: the team exerts no direct or indirect influence on the vehicle during the entire qualifying process
  - "Direct/indirect influence" includes:
    - Touching the vehicle
    - Operating the joystick
    - Touching the keyboard/mouse
    - Operating the computer
  - Recommended: keep hands away from joystick / laptop to demonstrate non-intervention



# Example of Qualifying Result

| Rank | Fully Autonomous | Q3 (fastest lap) | Q2 (laps) | Q1 (laps) |
|------|------------------|------------------|-----------|-----------|
| 1    | O                | 10.8 s           | 3         | 3         |
| 2    | O                | 11.2 s           | 3         | 3         |
| 3    | O                | 13.0 s           | 3         | 3         |
| 4    | X                | 9.9 s            | 3         | 3         |
| 5    | X                | 10.1 s           | 3         | 3         |
| 6    | X                | 12.2 s           | 3         | 3         |
| 7    | X                | 13.2 s           | 3         | 3         |
| 8    | X                | -                | 2         | 3         |
| 8    | X                | -                | 2         | 3         |
| 10   | X                | -                | 1         | 3         |
| 11   | X                | -                | 0         | 2         |
| 12   | X                | -                | 0         | 1         |
| 13   | X                | -                | 0         | 0         |
| 13   | X                | -                | 0         | 0         |

Assume 12 teams;  
3 teams pass each of Q1/Q2/Q3, and  
3 teams satisfy Fully Autonomous.

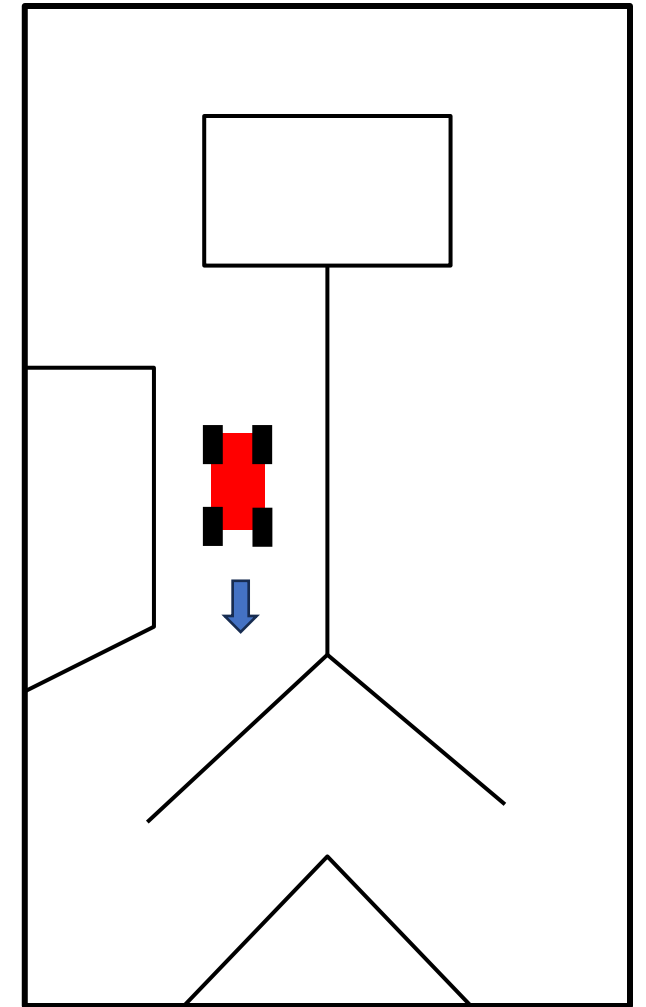
Fully Autonomous teams are ranked  
first by Q3 time, then Q3 passers,  
repeating down to Q1.



# Rules for Qualifying

- **Static Obstacles**

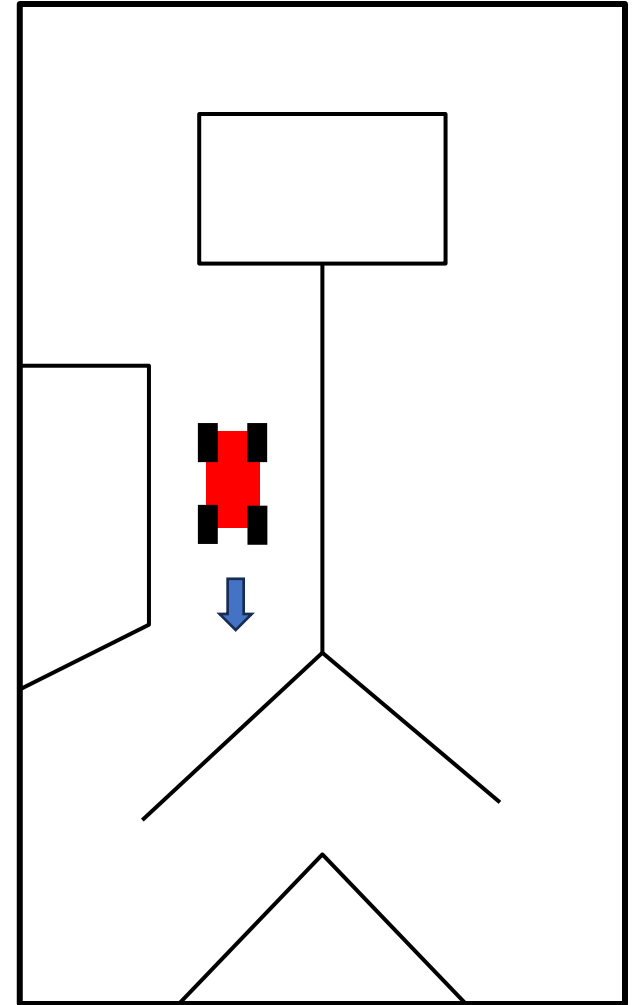
- The random obstacles placed in Q2 are static obstacles
- Each obstacle is smaller than  $0.5\text{m} \times 0.5\text{m}$
- Obstacle positions are placed randomly by the referee after Q1 is completed
- Obstacles are removed after Q2 is completed
- Obstacles are safely removed when vehicles are not affected
- There will be a **buffer lap** between each qualifying session (Q1→Q2 and Q2→Q3), but it will not count as a complete lap. During the buffer lap, you need to maintain autonomous mode.



# Rules for Qualifying

- **Record Invalidation**

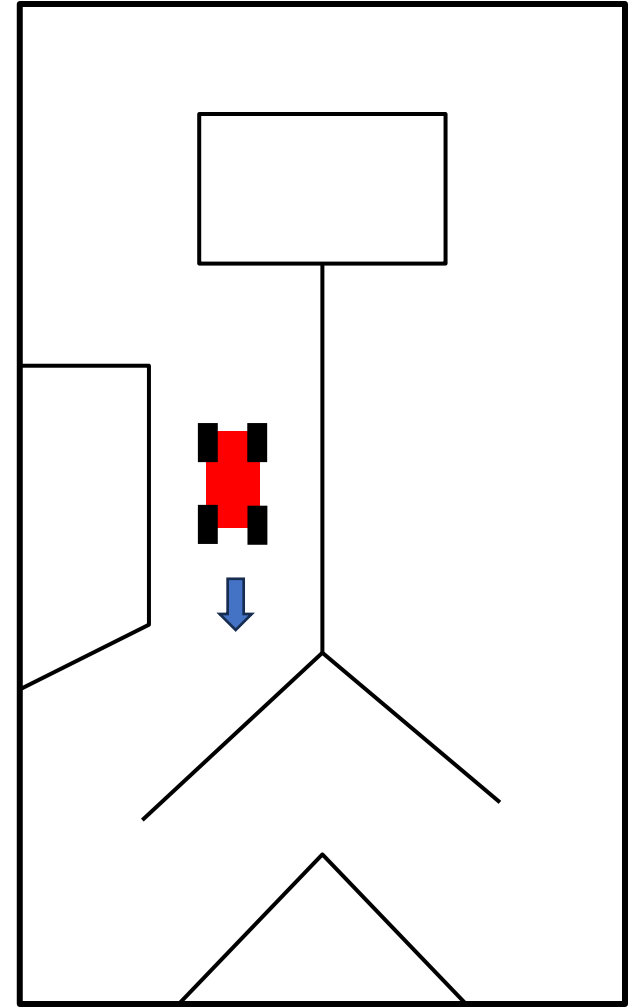
- Invalidates lap time and resets completed lap count if **human intervention** affects the vehicle
- Invalidates lap time and resets completed lap count if **contact with static obstacles** occurs
- If contact with track occurs but driving is possible without human intervention, it is considered minor contact and records remain valid
  - But racing track must be fixed



# Rules for Qualifying

- **Precautions**

- **All computation must be performed inside the vehicle!!**
- Manual (human) emergency brakes are strictly prohibited **during general driving**
- If a vehicle is removed from the track, **return it to the exit point**; only slight directional adjustment is allowed



# Penalty for Qualifying (One Rank Demotion)

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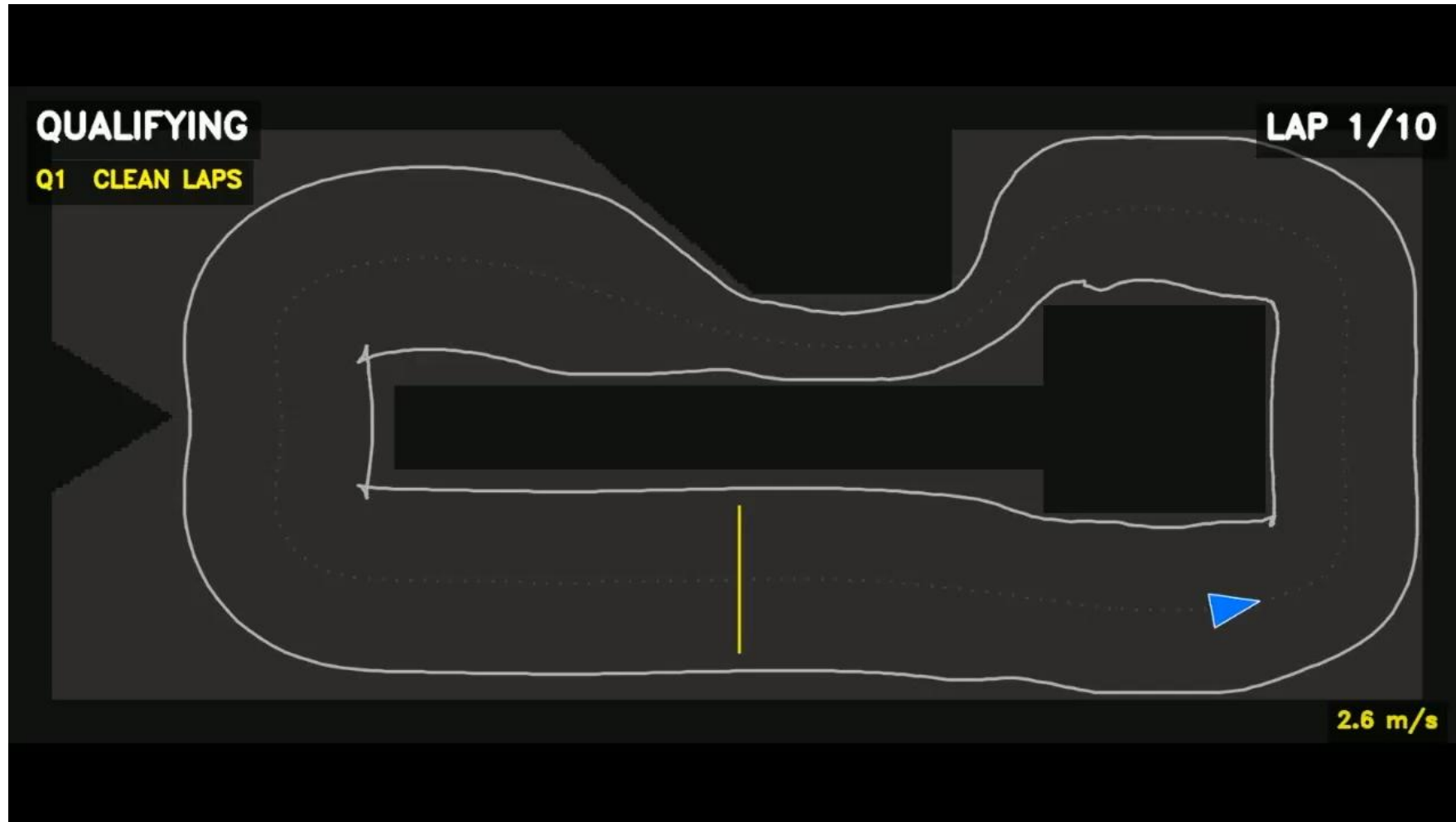
- **Using keyboard, mouse or Joystick during General Racing** (prohibited even if no data transmission occurs)
- **Do not generate or select modified paths manually based on obstacle positions**

# Penalty for Qualifying (One Rank Demotion)

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- **Exceptions:**
  - To transmit the start signal at start/restart
  - When the vehicle is not on the track
  - To stop after qualifying has ended
  - Transmit an initial guess for re-localization at stationary state.
  - When it need Emergency stop
    - when the referee declares a race stop
    - when driving is impossible due to a collision
    - when stuck on an obstacle for 5+ seconds
    - when the referee permits manual control

# Qualifying Demo



## Phase 4: Head-to-Head

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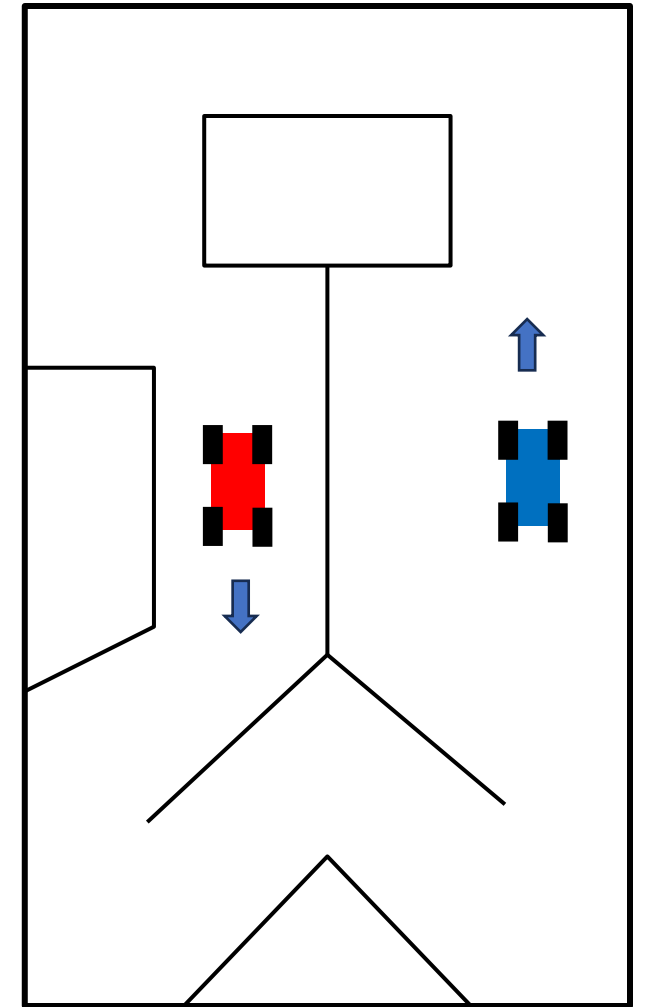
# Phase 4: Head-to-Head

- **Objective**

- **Complete 20 laps** first while avoiding obstacles and opponents
  - Races before the quarterfinals may be reduced to 10 laps

- **General**

- Both qualifying and head-to-head race use **same track**
- Two vehicles start from different starting lines positioned in opposite directions
- A total of **3 static obstacles** are used.
  - After all team complete race preparation, each team places one obstacle and the referee places one more
  - The track width must provide a **minimum clearance of 0.5 m**.
- Static obstacles are removed when the leading vehicle **completes the half of total laps**
- **Double elimination tournament**





# Phase 4: Head-to-Head

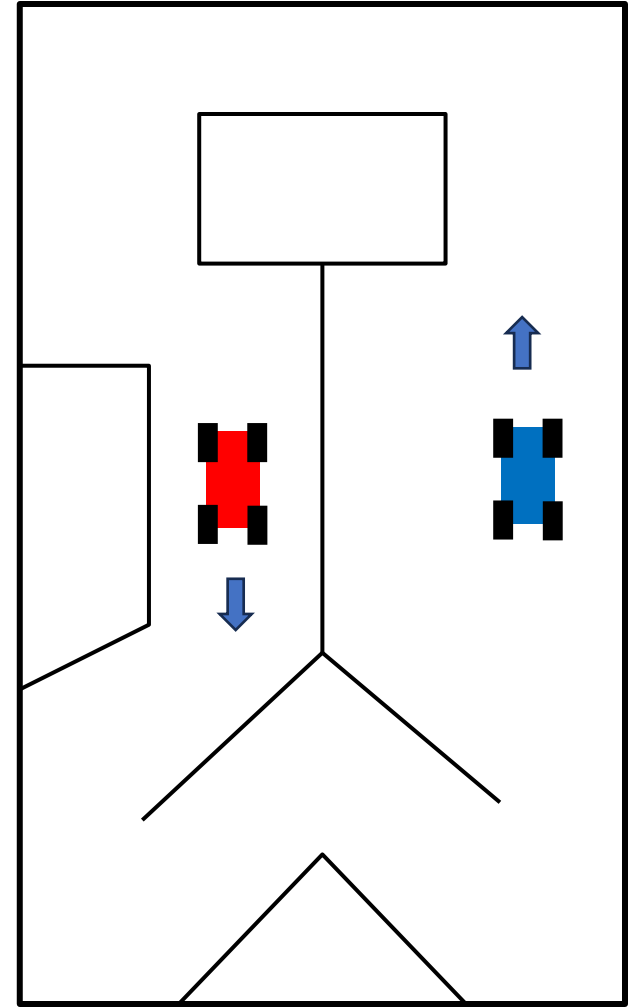
- **Collisions**

- **Track boundaries and static obstacles**

- Restore track and obstacles

- **Vehicle to vehicle**

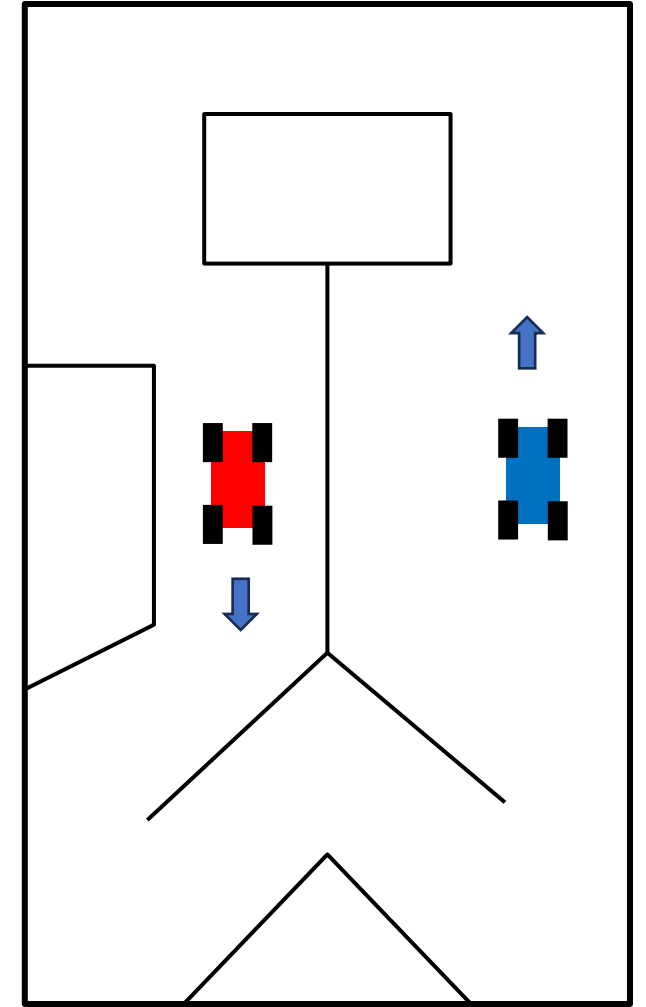
- When the victim is clearly decided by the referee, the race will be stopped if
      1. the victim car cannot continue to drive
      2. the victim car gets overtaken.



# Phase 4: Head-to-Head

- **Precautions**

- **Do not stop the race without the referee's stop signal!!**
- **All computation must be performed inside the vehicle!!**
- Data must not be transmitted to vehicles during the race.
- Use the emergency stop only for permitted situations.
- No protests regarding Wi-Fi will be accepted.
- Humans are not allowed on the track
  - All hardware repair/maintenance on the track is prohibited



# Rules for Head-to-Head (Warning)

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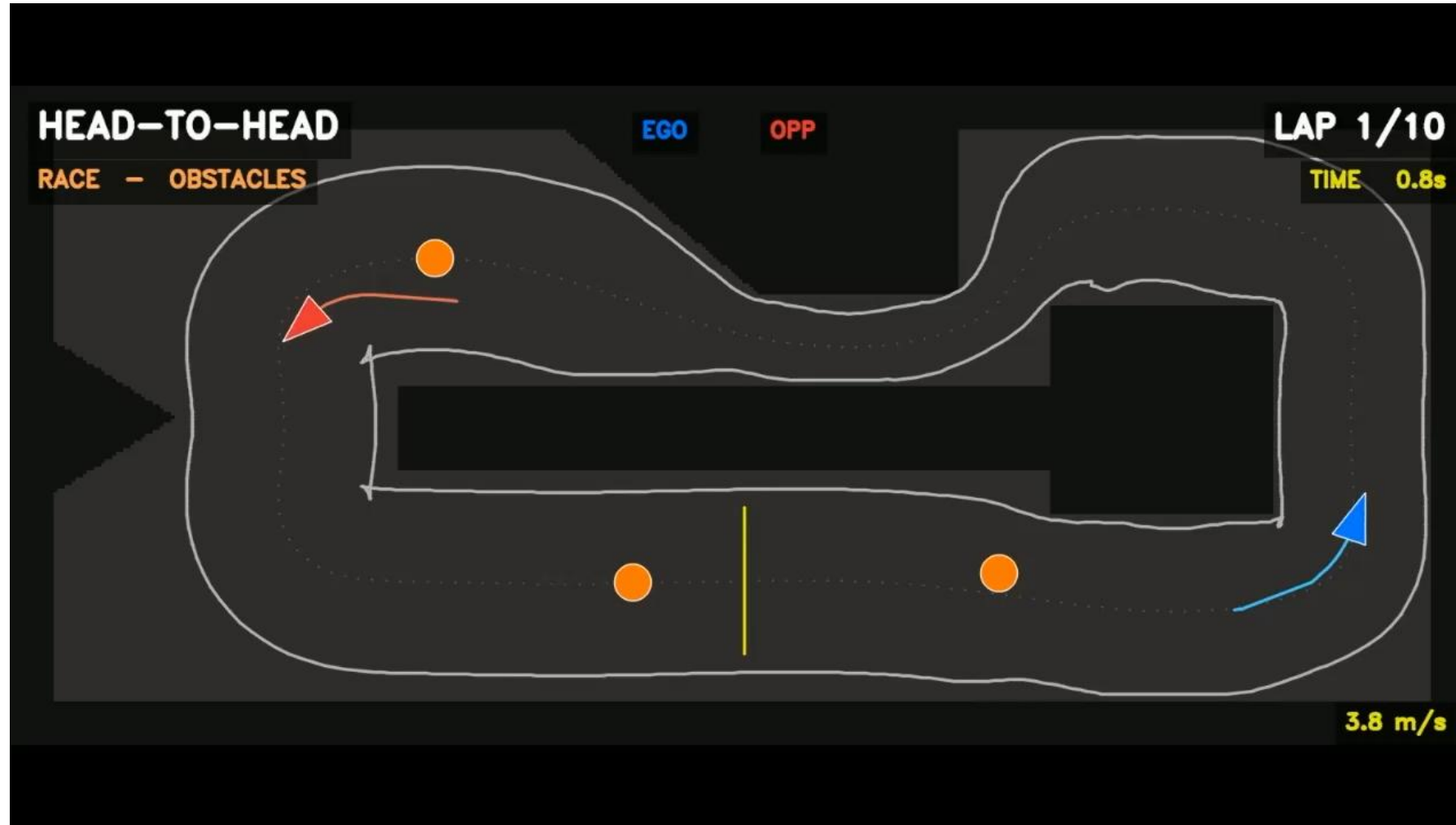
- Fault start
- Not actively restoring the track
- **Detection box not tightly fixed during the race**
- **Intervention with your own vehicle during racing**
  - Using keyboard, mouse or Joystick during general racing without referee's permission
- **A human's interference with the opponent vehicle during racing**
  - When going to fix the track and being detected by the opponent's detection module, affecting its driving

# Rules for Head-to-Head (Additional lap)

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- Accumulation of 3 warnings
- A human's fatal interference with the opponent vehicle during racing
  - When going to fix the track causes physical contact with the opponent vehicle affecting its driving
- A clear offending vehicle and victim vehicle exist, and the victim vehicle becomes unable to drive
- A complete rear-end collision with a large impact
- An accident with a large impact, even if not a complete rear-end collision
- **Disqualified with accumulation of 3 additional lap penalties**

# Head-to-Head Demo



# QnA

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Use Slack Channel(*#2026ifac-comeptition*) for additional questions!

[https://join.slack.com/t/robo-racer/shared\\_invite/zt-40e907kle-eJeqgl3\\_EsWi71T5UPAVuw](https://join.slack.com/t/robo-racer/shared_invite/zt-40e907kle-eJeqgl3_EsWi71T5UPAVuw)

